Sec: SR.IIT _*CO-SC(MODEL-A) **Date: 03-01-26**
Time: 3 Hrs **Max. Marks: 300**

03-01-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-7_ERRATA

REPLACED QUESTIONS

35. The number of neutrons present in the more abundant isotope of boron is 'x'. Amorphous boron upon heating with air forms a product, in which the oxidation state of boron is 'y'. The value of x + y is …

A) 4 B) 6 C) 3 D) 9

38. Which halogen is known to cause the reaction given below:

$2Cu^{2+} + 4X^{-} \rightarrow Cu_2X_2(s) + X_2$

A) Only Iodine B) Only Bromine C) All halogens D) Only Chlorine

39. The nature of oxides V_2O_3 and CrO is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is:

A) X = acidic Y = acidic
B) X = basic Y = amphoteric
C) X = basic Y = basic
D) X = amphoteric Y = basic

40. Which one of the following complexes will exhibit the least paramagnetic behaviour? [Atomic number, Cr = 24, Mn = 25, Fe = 26, Co = 27]

A) $\left[Cr(H_2O)_6\right]^{2+}$ B) $\left[Fe(H_2O)_6\right]^{2+}$ C) $\left[Co(H_2O)_6\right]^{2+}$ D) $\left[Mn(H_2O)_6\right]^{2+}$

41. The Crystal Field Stabilization Energy (CFSE) magnetic moment (spin-only) of an octahedral aqua complex of a metal ion (M^{z+}) are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify (M^{Z+}):

A) V^{3+} B) Co^{2+} C) Cr^{3+} D) Mn^{4+}

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46. The vapour pressure of pure benzene and methyl benzene at 27^0 C is given as 80 Torr and 24 Torr, respectively. The mole fraction of methyl benzene in vapour phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is ___ $\times 10^{-2}$ (nearest integer)

48. A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B(in s) is_____ (Use ln 2 = 0.693)

49. Number of oxygen atoms present in chemical formula of fuming sulphuric acid is_____

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